# A Control-Theoretic Perspective on Modern Optimizers

## Optimizer Dynamics, Exact Nonlinear Responses, and Feedback Controller Synthesis

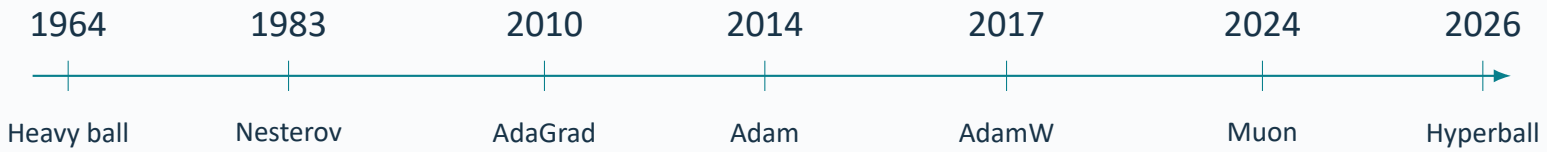
AI Safety PhD Group Meeting

60-minute paper sharing

History (21 min) Control (12 min) Theory (16 min) Case study (9 min)

Literature checked on 9 November 2026

# Optimizer history: several design branches




Temporal memory: momentum, EMA mixtures
Coordinate scaling: AdaGrad, Adam, AdamW
Matrix structure: Shampoo, SOAP, Muon
State constraints: Hyperball, SSO

Dates indicate the cited public work. Formal publication may occur later.

Polyak (1964 journal); Nesterov (1983 journal); Duchi, Hazan, and Singer (2010 public; JMLR 2011); Kingma and Ba (2014 public; ICLR 2015); Loshchilov and Hutter (2017 public; ICLR 2019); Jordan (2024 author write-up); Wen, Dang, Lyu, Ma, and Liang (2026 preprint, v2)

# What an optimizer changes

$$\min_W f(W), \qquad g_t = \nabla f(W_t)$$
$$s_t = F(s_{t-1}, g_t), \qquad W_{t+1} = W_t - \eta_t u_t$$

- The loss supplies a gradient. The optimizer carries internal state $s_t$.
- Training cost includes the number of steps and the cost of each step.
- For safety research, numerical stability is one training property. Behavioral safety needs separate evaluation.

Notation introduced in this presentation

# Gradient descent

$$W_{t+1} = W_t - \eta \nabla f(W_t)$$

$$f(x) = \tfrac{1}{2}(x_1^2 + 25x_2^2), \qquad \nabla f(x) = (x_1, 25x_2)$$

- One scalar step size scales every coordinate.
- The same step can be slow along $x_1$ and unstable along $x_2$.
- This quadratic is an exactly defined example, rather than a local approximation.

**Illustrative exact system**. Defined simulation: $f(x) = \frac{1}{2}(x_1^2 + 25x_2^2)$

# SGD and mini-batch gradients

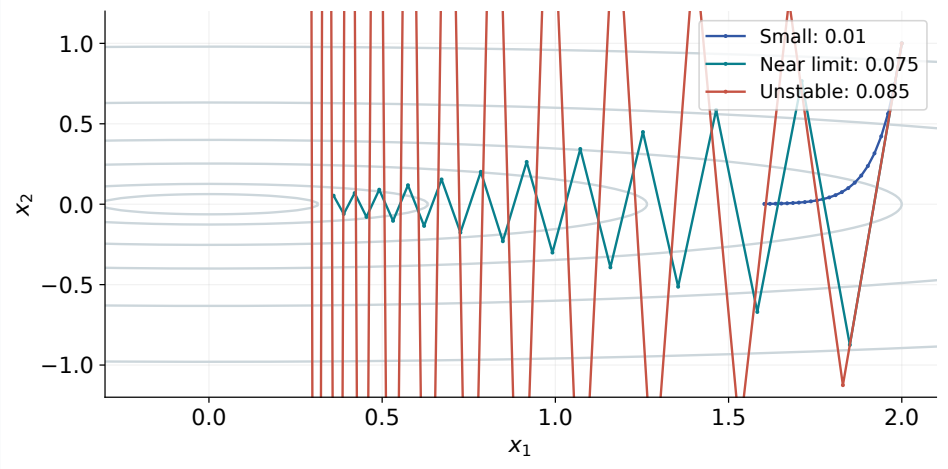
$$g_t = \frac{1}{|\mathcal{B}_t|} \sum_{i \in \mathcal{B}_t} \nabla \ell_i(W_t)$$

$$g_t = \nabla f(W_t) + \xi_t$$

- Mini-batches reduce gradient computation per step.
- Unbiasedness requires assumptions about sampling.
- Noise can interact with optimizer memory and nonlinear normalization.

Hinton, Srivastava, and Swersky (2012 course); Kingma and Ba (2014 public; ICLR 2015)

# The learning rate controls the response




$$f(x) = \tfrac{1}{2}(x_1^2 + 25x_2^2), \qquad 0 < \eta < 2/25$$

**Derived in this presentation**. Analytical calculation and reproducible simulation

# Heavy-ball momentum adds a state

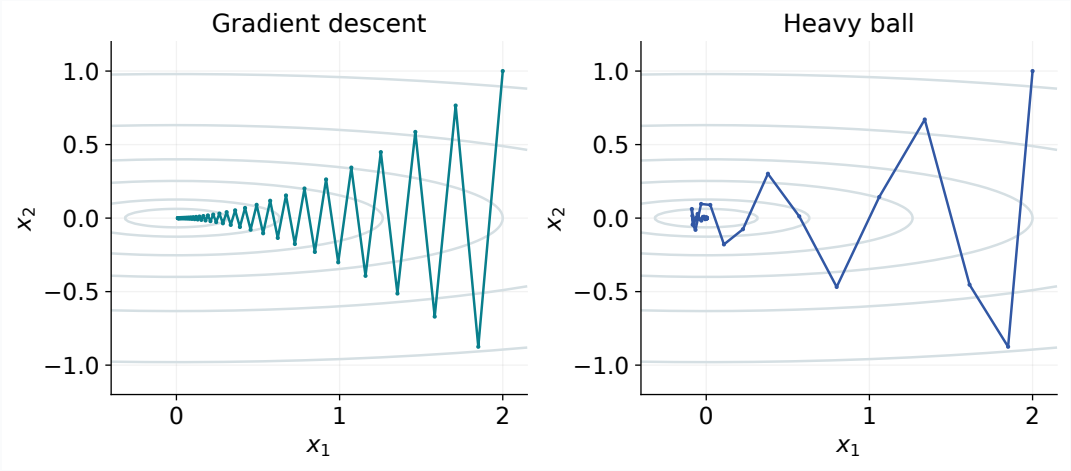
$$b_t = \beta b_{t-1} + g_t, \qquad W_{t+1} = W_t - \eta b_t$$

$$W_{t+1} = W_t - \eta g_t + \beta(W_t - W_{t-1})$$

- Persistent gradient components accumulate.
- Alternating components partially cancel.
- The extra state can also produce overshoot.

Polyak (1964 journal)

# Momentum on a defined quadratic




Same $f$, $x_0$, and $\eta = 0.075$. Heavy ball uses $\beta = 0.65$.

**Illustrative exact system**. Reproducible deterministic simulation

# Nesterov evaluates a look-ahead point

$$y_t = W_t + \beta_t(W_t - W_{t-1})$$

$$W_{t+1} = y_t - \eta \nabla f(y_t)$$

- The gradient evaluation point differs from heavy ball.
- Classical acceleration guarantees require objective and parameter assumptions.
- Muon uses a Nesterov-style gradient blend, specified separately later.

Nesterov (1983 journal)

# Coordinate scales motivate adaptation

$$u_t = D_t g_t, \qquad D_t \text{ diagonal and history-dependent}$$

| Design | Information | Main limitation |
|---|---|---|
| Scalar step | One global scale | Unequal coordinate scales |
| Diagonal scaling | Coordinate statistics | No cross-coordinate mixing |
| Matrix scaling | Structured correlations | More state and computation |

Duchi, Hazan, and Singer (2010 public; JMLR 2011); Gupta, Koren, and Singer (2018 public; ICML 2018)

# AdaGrad accumulates squared gradients

$$a_t = a_{t-1} + g_t^{\odot 2}, \qquad u_t = \frac{g_t}{\sqrt{a_t} + \epsilon}$$

- Rarely active coordinates can retain larger effective steps.
- The accumulator never decreases.
- Long training histories can make effective steps very small.

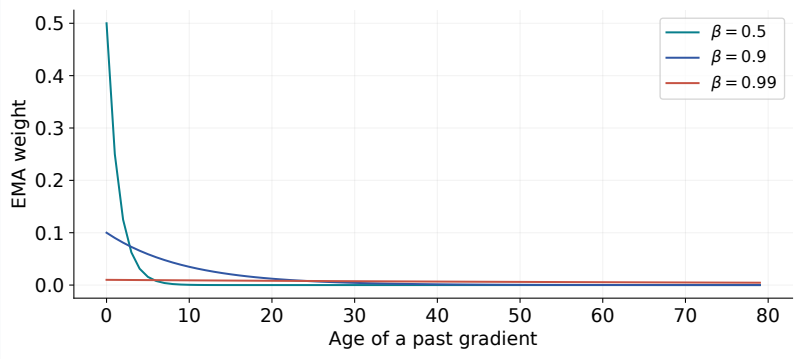
Duchi, Hazan, and Singer (2010 public; JMLR 2011)

# RMSProp forgets old squared gradients

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^{\odot 2}$$

$$u_t = \frac{g_t}{\sqrt{v_t} + \epsilon}$$

The denominator tracks a moving scale.




Tieleman and Hinton (RMSProp, 2012 course)

# Adam combines two moving averages

$$
\begin{aligned}
m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\
v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^{\odot 2}, \\
\widehat{m}_t &= m_t / (1-\beta_1^t), \quad \widehat{v}_t = v_t / (1-\beta_2^t), \\
u_t &= \widehat{m}_t / (\sqrt{\widehat{v}_t} + \epsilon).
\end{aligned}
$$

$m_0 = v_0 = 0$. Square, square root, and division are elementwise.

Kingma and Ba (2014 public; ICLR 2015)

# Bias correction removes zero-start attenuation

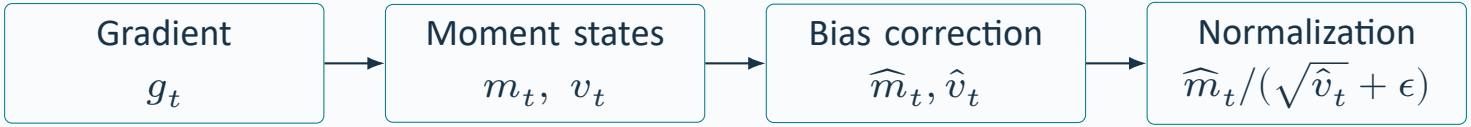
$$g_t = g \implies m_t = (1 - \beta_1^t)g$$

$$\widehat{m}_t = g, \qquad \hat{v}_t = g^{\odot 2}$$

Under a constant input, bias correction removes the initialization transient exactly.

**Derived in this presentation**. Kingma and Ba (2014 public; ICLR 2015); geometric-series derivation

# Adam separates memory and normalization




$$g = (2, 0.02) \quad \Longrightarrow \quad u_1 \approx (1, 1)$$

With zero initialization and negligible $\epsilon$, the first update equalizes coordinate magnitudes.

Kingma and Ba (2014 public; ICLR 2015)

# L2 regularization changes the adaptive input

Coupled: $u_t = D_t(g_t + \lambda W_t)$

Decoupled: $W_{t+1} = (1 - \eta_t \lambda) W_t - \eta_t D_t g_t$

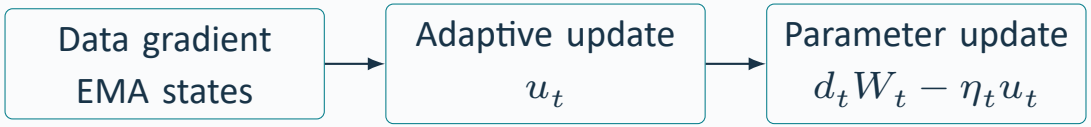
With adaptive $D_t$, the coupled penalty receives coordinate-dependent scaling.

In Adam, coupling also changes the moment states themselves.

Loshchilov and Hutter (2017 public; ICLR 2019)

# AdamW decouples the radial shrinkage

$$W_{t+1} = d_t W_t - \eta_t \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon}, \qquad d_t = 1 - \eta_t \lambda$$




Public preprint: 2017. Formal publication: ICLR 2019.

Loshchilov and Hutter (2017 public; ICLR 2019)

# AMSGrad changes the denominator memory

$$\overline{v}_t = \max(\overline{v}_{t-1}, v_t)$$

$$W_{t+1} = W_t - \eta_t \frac{m_t}{\sqrt{\overline{v}_t} + \epsilon}$$

- Maximum is elementwise. This is a simplified, uncorrected display convention.
- A persistent maximum changes the effective learning-rate sequence.
- The paper gives Adam counterexamples and qualified AMSGrad guarantees.

Reddi, Kale, and Kumar (ICLR 2018)

# Lion and GaLore target optimizer state cost

| Method | Main design | Public / formal |
|---|---|---|
| Lion | Sign of a momentum blend. One full moment state. | 2023 / NeurIPS 2023 |
| GaLore | Project gradients into a changing low-rank space. Keep full parameters. | 2024 / ICML 2024 |

Projection and sign operations introduce nonlinearities beyond temporal filtering.

Chen et al. (2023 public; NeurIPS 2023); Zhao et al. (2024 public; ICML 2024)

# Newton and L-BFGS use curvature information

Newton: $u_t = \nabla^2 f(W_t)^{-1} \nabla f(W_t)$

L-BFGS: $u_t = H_t g_t, \quad H_t$ from recent $(s_i, y_i)$

- Newton needs an invertible Hessian and safeguards in nonconvex problems.
- Limited-memory quasi-Newton methods avoid storing a dense inverse.
- Original limited-storage work: 1980. L-BFGS study: 1989.

Nocedal; Liu and Nocedal (1980 limited storage; 1989 L-BFGS)

# Natural gradient uses a statistical metric

$$F(W) = \mathbb{E}\big[\nabla_W \log p_W \, \nabla_W \log p_W^\top\big]$$

$$u_t = (F(W_t) + \delta I)^{-1} g_t$$

- The Fisher matrix describes changes in the model distribution.
- Damping makes the displayed inverse well-defined when $\delta > 0$.
- The Fisher and the loss Hessian coincide only in special settings.

Amari (Neural Computation 1998)

# Shampoo preserves tensor structure

$$L_t = \delta I + \sum_{i=1}^{t} G_i G_i^\top, \qquad R_t = \delta I + \sum_{i=1}^{t} G_i^\top G_i$$

$$U_t = L_t^{-1/4} G_t R_t^{-1/4}$$

- Displayed form: original matrix Shampoo, without grafting or momentum.
- Left and right statistics couple coordinates within a matrix.
- Public preprint and ICML publication: 2018.

Gupta, Koren, and Singer (2018 public; ICML 2018)

# Sophia combines curvature scaling and clipping

$$U_t = \mathrm{clip}\left(\frac{m_t}{\max(\gamma h_t, \epsilon)}, -1, 1\right)$$

- $h_t$: moving estimate of diagonal curvature, refreshed periodically.
- Clipping bounds each displayed update coordinate.
- Public: May 2023. Formal: ICLR 2024.

Liu, Li, Hall, Liang, and Ma (2023 public; ICLR 2024)

# SOAP runs Adam in a changing matrix basis

$$\widetilde{G}_t = Q_{L,t}^\top G_t Q_{R,t}$$

$$\widetilde{U}_t = \mathsf{AdamStateUpdate}(\widetilde{G}_t), \qquad U_t = Q_{L,t} \widetilde{U}_t Q_{R,t}^\top$$

The $Q$ matrices come from Shampoo statistics. State handling across basis refreshes is part of the algorithm.

Public: September 2024. Formal: ICLR 2025.

Vyas et al. (2024 public; ICLR 2025)

# Memory and computation depend on the variant

| Method on $m \times n$ | Representative persistent state | Main extra work |
|---|---|---|
| AdamW | $2mn$ | Elementwise operations |
| Muon | $mn$ | Five NS matrix iterations |
| Shampoo | $m^2 + n^2$ | Matrix inverse roots |
| SOAP | $2mn + O(m^2 + n^2)$ | Basis refresh and rotations |

Counts exclude parameters, gradients, master copies, sharding, and work buffers. Blocking changes Shampoo and SOAP costs.

Kingma and Ba (2014 public; ICLR 2015); Gupta, Koren, and Singer (2018 public; ICML 2018); Vyas et al. (2024 public; ICLR 2025); Jordan and contributors (Muon code, f90a42b)

# Muon shapes matrix updates

$$B_t = \mu B_{t-1} + (1-\mu)G_t$$

$$Z_t = (1-\mu)G_t + \mu B_t, \qquad O_t = \mathcal{N}_5(Z_t)$$

- Orthogonalization follows a Nesterov-style temporal filter.
- Use matrix updates for hidden weights and a separate optimizer for other parameters.
- Public design write-up: December 2024.

Jordan (2024 author write-up); Jordan and contributors (Muon code, f90a42b)

# The polar factor explains the spectral idea

$$Z = U\Sigma V^\top, \qquad \text{polar}(Z) = UV^\top$$

- For full rank, all nonzero singular values become one.
- A finite polynomial maps singular values without computing an SVD.
- At rank deficiency, the polar factor needs a convention. Polynomial maps keep zero singular values at zero.

Jordan (2024 author write-up); Jordan and contributors (Muon code, f90a42b)

# Muon uses five finite polynomial iterations

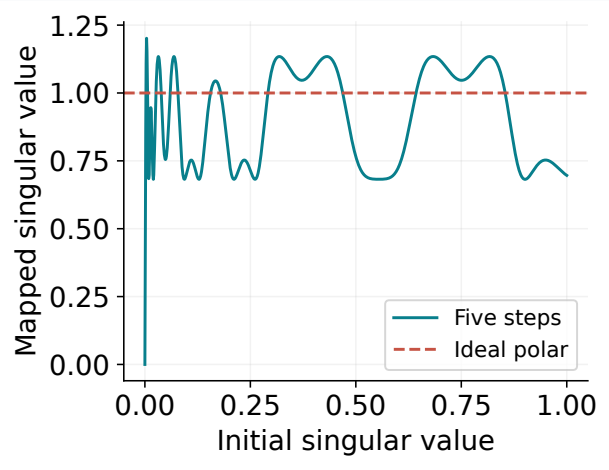
$$X_0 = Z/(\|Z\|_F + \epsilon_{\text{NS}})$$

$$A_k = X_k X_k^\top,$$

$$X_{k+1} = aX_k + (bA_k + cA_k^2)X_k,$$

$(a, b, c) = (3.4445, -4.775, 2.0315)$

$k = 0, \ldots, 4$. Transpose tall inputs.




Jordan (2024 author write-up); Jordan and contributors (Muon code, f90a42b)

# Recent work revisits temporal memory

| Method | Main idea | Public / formal |
|---|---|---|
| AdEMAMix | Combine fast and slow first-moment EMAs. | 2024 / ICLR 2025 |
| MARS | Combine preconditioning with recursive variance reduction. | 2024 / not established here |

Changing a history filter can change the response even when the update direction remains coordinatewise.

Pagliardini, Ablin, and Grangier (2024 public; ICLR 2025); Yuan et al. (2024 public; arXiv v4)

# Recent work revisits the matrix spectrum

| Method | Main idea | Public / formal |
|---|---|---|
| Dion | Amortized orthonormalization with sharding-aware updates. | 2025 / not established here |
| COSMOS | SOAP in a leading subspace, Muon in the residual. | 2025 / ICLR 2026 |
| Spectra | Suppress dominant spectral spikes while protecting the tail. | 2026 / ICML 2026 |

Spectra here denotes Huang et al., arXiv:2602.11185, not other similarly named methods.

Ahn et al. (2025 public; arXiv v3); Liu et al. (2025 public; ICLR 2026); Huang et al. (2026 public; ICML 2026)

# State constraints and evidence at scale

| Method | Main idea | Public / formal |
|---|---|---|
| Hyperball | Fix matrix Frobenius radius and trial-step scale. | 2026 / preprint cited |
| SSO | Constrain weights and updates in spectral norm. | 2026 / ICML 2026 |
| SCALE | Column normalization plus output-layer momentum. | 2025 / ICML 2026 |

Evidence question: matched tuning, training horizon, numerical precision, and wall-clock cost can change the ranking.

Wen, Dang, Lyu, Ma, and Liang (2026 preprint, v2); Xie et al. (2026 public; ICML 2026); Glentis, Li, Han, and Hong (2025 public; ICML 2026); Khona et al. (2026 preprint); Xiao et al. (2026 preprint)

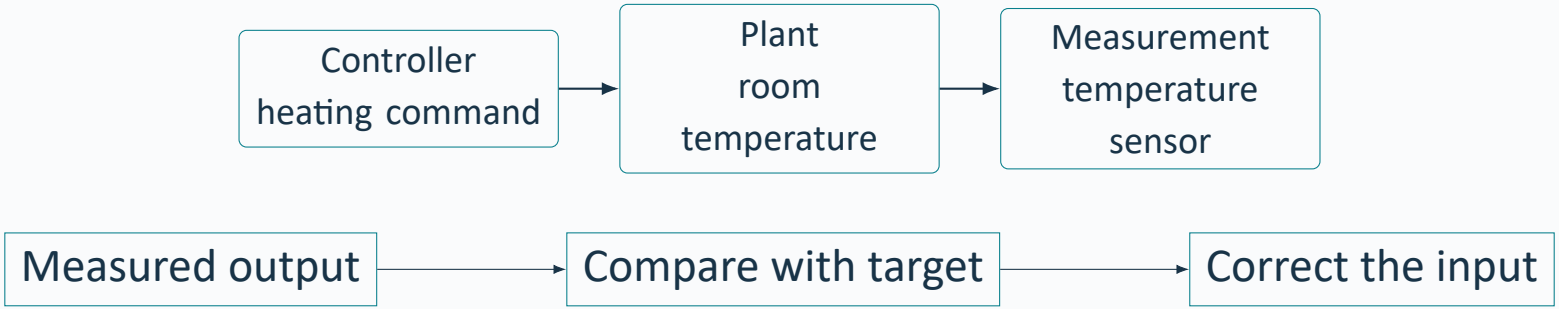
# Why dynamics matter for optimization

$$\text{same } g_t \quad + \quad \text{different internal states} \quad \Longrightarrow \quad \text{different updates}$$

- A training trajectory depends on memory, feedback, and disturbances.
- Control theory describes transient response and robustness.
- A certificate requires a specified system class and assumptions.

Hu and Lessard (ACC 2017); Lessard, Recht, and Packard (2014 public; SIAM 2016)

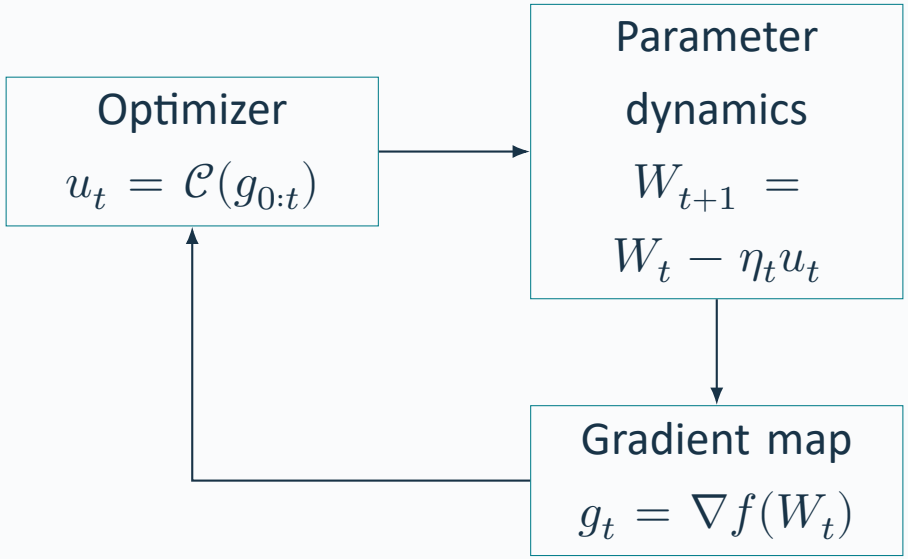
# Feedback control




A feedback controller changes its action based on the observed state or output.

Teaching example based on standard feedback concepts; Hu and Lessard (ACC 2017)

# Optimization as a closed loop




The optimizer is a controller with memory. The gradient map closes the loop.

Hu and Lessard (ACC 2017); diagram redrawn for this presentation

# State-space models keep the memory explicit

$$x_{t+1} = Ax_t + Bu_t, \qquad y_t = Cx_t + Du_t$$

$$x_{t+1} = F(x_t, u_t) \quad \text{for a nonlinear system}$$

- The state contains everything required to predict the next step.
- For momentum, include the parameter and momentum buffer.
- Adam additionally needs the second moment and step counter.

Standard state-space notation; Chakrabarti and Chopra (2022 public; arXiv v2 2023)

# The Z-transform turns delays into factors

$$X(z) = \sum_{t=0}^{\infty} x_t z^{-t}, \qquad \mathcal{Z}\{x_{t-1}\} = z^{-1}X(z)$$

$$y_t = \beta y_{t-1} + (1-\beta)u_t \quad \Longrightarrow \quad H(z) = \frac{1-\beta}{1-\beta z^{-1}}$$

Transfer functions describe zero-state responses. Nonzero initial conditions contribute separate terms.

**Derived in this presentation**. Hu and Lessard (ACC 2017); transform calculation

# The parameter integrator includes a minus sign

$$x_{t+1} = x_t - u_t$$

$$z[X(z) - x_0] = X(z) - U(z)$$

$$X(z) = \frac{zx_0}{z-1} - \frac{1}{z-1}U(z)$$

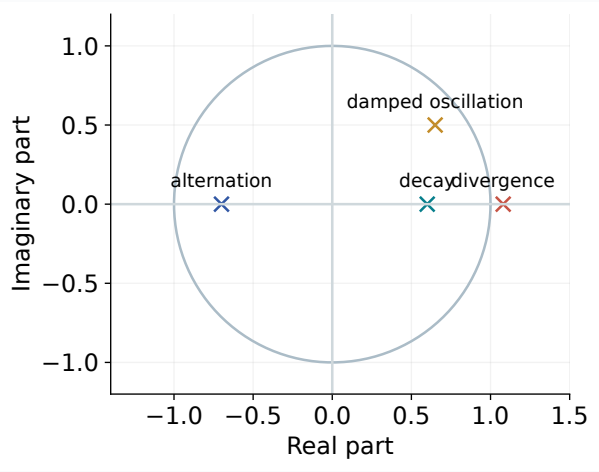
Zero-state map: $X/U = -1/(z-1)$. Equivalently, use a positive integrator with an external negative-feedback junction.

**Derived in this presentation**. Exact transform under the stated indexing

# Poles determine linear asymptotic behavior

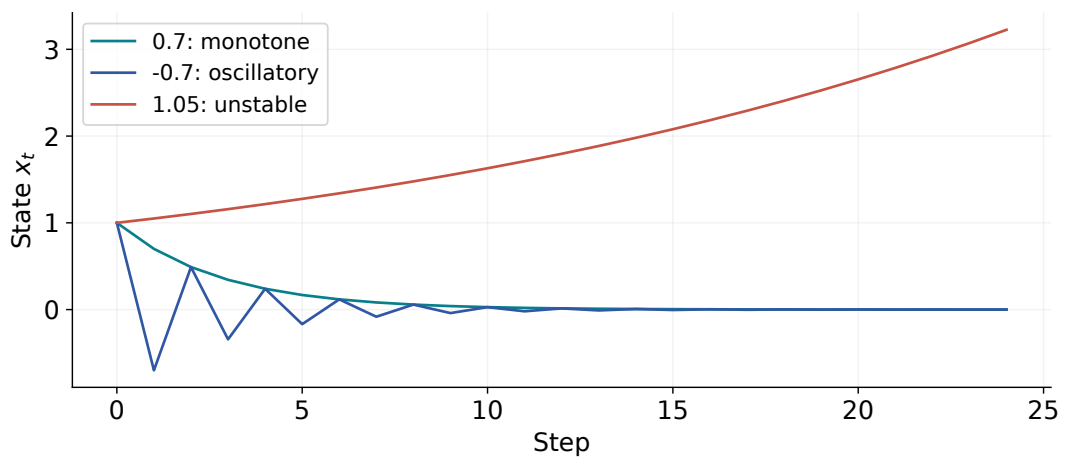
$$x_{t+1} = ax_t \quad \Longrightarrow \quad x_t = a^t x_0$$

- $|a| < 1$: asymptotic decay.
- $a < 0$: sign alternation.
- For a finite-dimensional LTI state model, all eigenvalues must lie inside the unit disk.




Exact scalar system; standard discrete-time stability criterion

# Transient response describes the route to equilibrium




Rise time, overshoot, oscillation, and settling time characterize different parts of a response.

**Illustrative exact system**. Defined scalar recurrences

# Steady-state error depends on the closed loop

$$f(x) = \tfrac{h}{2}x^2, \qquad g_t = hx_t + \xi$$

$$x_{t+1} = (1 - \eta h)x_t - \eta\xi$$

$$|1 - \eta h| < 1 \quad \Longrightarrow \quad x_\infty = -\xi/h$$

Parameter integration alone does not remove a persistent bias in the measured gradient.

**Derived in this presentation**. Exact quadratic with constant gradient bias

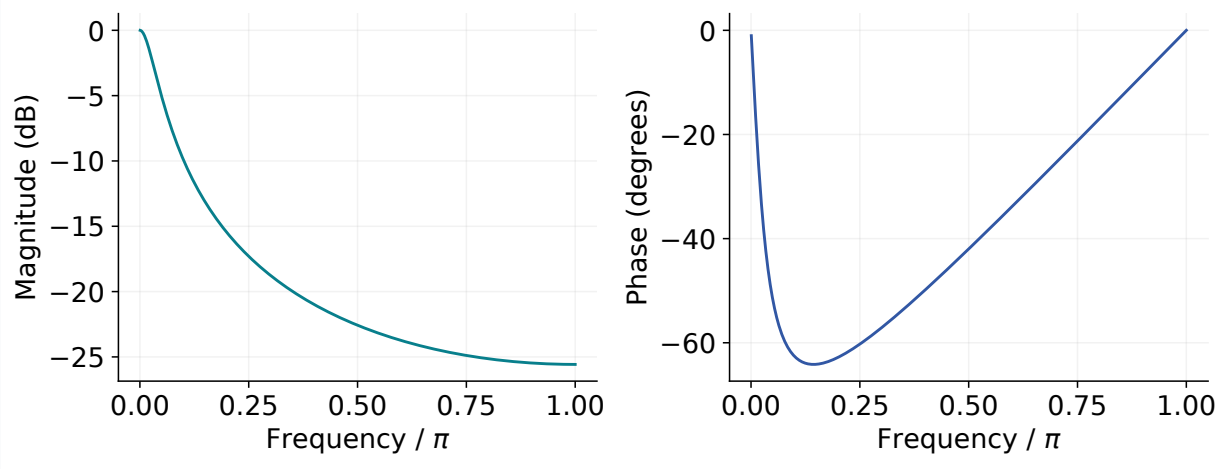
# Frequency response requires a specified input

$$u_t = \mathrm{Re}(ae^{j\omega t}) \quad \Longrightarrow \quad y_t^* = \mathrm{Re}(H(e^{j\omega})ae^{j\omega t})$$

- Magnitude describes sinusoidal amplitude scaling.
- Phase describes temporal lag or lead.
- A controller frequency response and a training-loop response have different inputs and outputs.

LTI steady-state calculation; Hu and Lessard (ACC 2017)

# A Bode plot of the EMA filter




$$H(z) = \frac{1-\beta}{1-\beta z^{-1}}, \qquad \beta = 0.9$$

**Derived in this presentation**. Analytical frequency response, not a training experiment

# Disturbance rejection is a closed-loop property

$$\frac{X(z)}{\Xi(z)} = -\frac{C(z)}{z - 1 + hC(z)} \qquad (g_t = hx_t + \xi_t)$$

$$\|H\|_\infty = \sup_\omega \sigma_{\max}(H(e^{j\omega})), \quad \|H\|_2^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} \|H(e^{j\omega})\|_F^2 d\omega$$

- For stable LTI maps: worst-case energy amplification versus white-noise variance gain.
- A small-gain certificate also needs a bounded uncertainty operator and a well-posed loop.

Hu and Lessard (ACC 2017)

# Modern optimizer nonlinearities remain explicit

| Operation | Why one global LTI map fails |
|---|---|
| $g \mapsto g^{\odot 2}$ | Nonlinear scaling and frequency mixing |
| $(m, v) \mapsto m/(\sqrt{v} + \epsilon)$ | State-dependent division |
| $Z \mapsto Z/(\|Z\|_F + \epsilon)$ | Normalization couples entries |
| $X \mapsto aX + bXX^\top X + c(XX^\top)^2 X$ | Matrix polynomial nonlinearity |

Time-dependent bias correction and schedules add time variation.

Kingma and Ba (2014 public; ICLR 2015); Jordan and contributors (Muon code, f90a42b)

# Lyapunov functions and dissipativity

$$V(x_{t+1}) - V(x_t) \leq -\alpha \|x_t\|^2$$

$$V(x_{t+1}) - V(x_t) \leq s(u_t, y_t)$$

- A positive storage function quantifies the state.
- A supply rate accounts for the input-output exchange.
- Strict decay, boundedness, and convergence require different hypotheses.

Hu and Lessard (ICML 2017)

# IQCs constrain an operator without linearizing it

$$y = \Delta(u), \qquad z = \Psi \begin{bmatrix} u \\ y \end{bmatrix}$$

$$\sum_{t=0}^{T} z_t^\top M z_t \geq 0 \qquad \text{for every } T \geq 0$$

- This displayed form is a hard, time-domain IQC.
- A filter $\Psi$ can encode memory in the constraint.
- A valid constraint alone does not certify the interconnected system.

Lessard, Recht, and Packard (2014 public; SIAM 2016)

# A toolbox with clear roles

| Tool | What it contributes |
|---|---|
| State space | Exact internal memory and timing |
| Z-transform | Linear filters and exact zero-state responses |
| Nonlinear recurrences | Complete update maps and periodic responses |
| Dissipativity and IQCs | Conditional input-output and stability certificates |

The next chapter preserves the nonlinear optimizer and analyzes the parts that admit exact statements.

Lessard, Recht, and Packard (2014 public; SIAM 2016); Hu and Lessard (ACC 2017); Hu and Lessard (ICML 2017)

# Control and optimization: established foundations

| Work | Public / formal | Scope |
|---|---|---|
| Lessard et al. | 2014 / 2016 | IQCs and numerical rate certificates |
| Hu and Lessard | 2017 / 2017 | Loop shaping and small gain |
| Hu and Lessard | 2017 / 2017 | Dissipativity for Nesterov |
| Chakrabarti and Chopra | 2022 / unverified | Adaptive state-space models and AdamSSM |
| Chen et al. | 2024 / 2024 | Continuous-time PID optimizer and discretizations |

Verified formal venues: SIAM J. Optim., ACC, ICML, Nature Communications.

Lessard, Recht, and Packard (2014 public; SIAM 2016); Hu and Lessard (ACC 2017); Hu and Lessard (ICML 2017); Chakrabarti and Chopra (2022 public; arXiv v2 2023); Chen et al. (Nature Communications 2024)

# SGD is proportional feedback on a quadratic

$$u_t = \eta g_t, \quad x_{t+1} = x_t - u_t, \quad g_t = h x_t$$

$$z - 1 + \eta h = 0 \quad \Longrightarrow \quad z_{\text{cl}} = 1 - \eta h$$

$$h > 0: \qquad |z_{\text{cl}}| < 1 \iff 0 < \eta h < 2$$

Here the controller includes the step size. The objective is exactly $f(x) = hx^2/2$.

**Derived in this presentation**. Hu and Lessard (ACC 2017); algebra shown here

# Momentum changes the closed-loop polynomial

$$C_{\mathsf{HB}}(z) = \frac{\eta}{1 - \beta z^{-1}}$$

$$z^2 - (1 + \beta - \eta h)z + \beta = 0$$

$$0 \le \beta < 1, \quad h > 0 : \qquad 0 < \eta h < 2(1 + \beta)$$

The last condition follows from the second-order Jury test for this exact scalar quadratic.

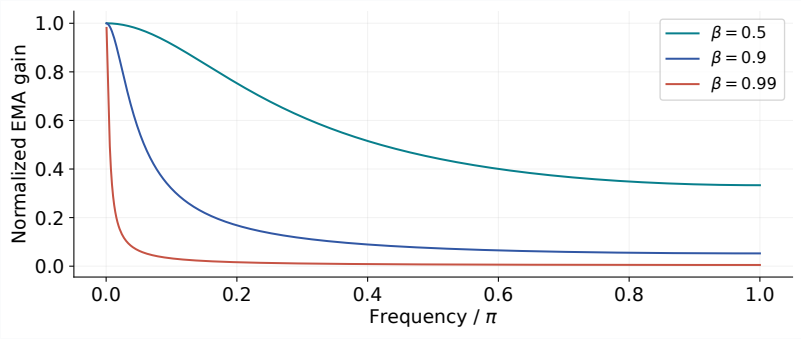
**Derived in this presentation**. Hu and Lessard (ACC 2017); Jury calculation in Appendix A2

# Momentum filters temporal gradient variation

$$H_{\text{EMA}}(1) = 1$$

$$H_{\text{EMA}}(-1) = \frac{1-\beta}{1+\beta}$$

Unnormalized heavy ball has an extra factor $1/(1-\beta)$.




**Derived in this presentation**. Exact filter responses

# A complete optimizer has no single global Bode plot

$$g \longmapsto (m, v) \longmapsto \frac{\widehat{m}}{\sqrt{\widehat{v}} + \epsilon}$$
$$G \longmapsto Z \longmapsto \mathcal{N}_5(Z)$$

- The linear filters admit transfer functions.
- The nonlinear outputs can create harmonics and amplitude dependence.
- The training loop also contains $ablaf$, schedules, and weight decay.

Kingma and Ba (2014 public; ICLR 2015); Jordan and contributors (Muon code, f90a42b)

# An exact nonlinear feedback representation

$$
\begin{aligned}
g_t &= \nabla f(W_t), \\
s_t &= F(s_{t-1}, g_t), \\
u_t &= \Phi(s_t, g_t, W_t, t), \\
W_{t+1} &= W_t - \eta_t u_t.
\end{aligned}
$$

Our indexing: observe $W_t$, read $g_t$, update the controller state, then write $W_{t+1}$.

**Control-theoretic reinterpretation**. Exact modeling convention for this presentation

# AdamW: the first-moment filter is linear

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t, \qquad m_0 = 0$$

$$\frac{M(z)}{G(z)} = \frac{1-\beta_1}{1-\beta_1 z^{-1}}$$

$$\widehat{m}_t = \frac{m_t}{1-\beta_1^t}$$

Bias correction is an exact time-dependent output gain.

**Derived in this presentation**. Kingma and Ba (2014 public; ICLR 2015); transform derivation

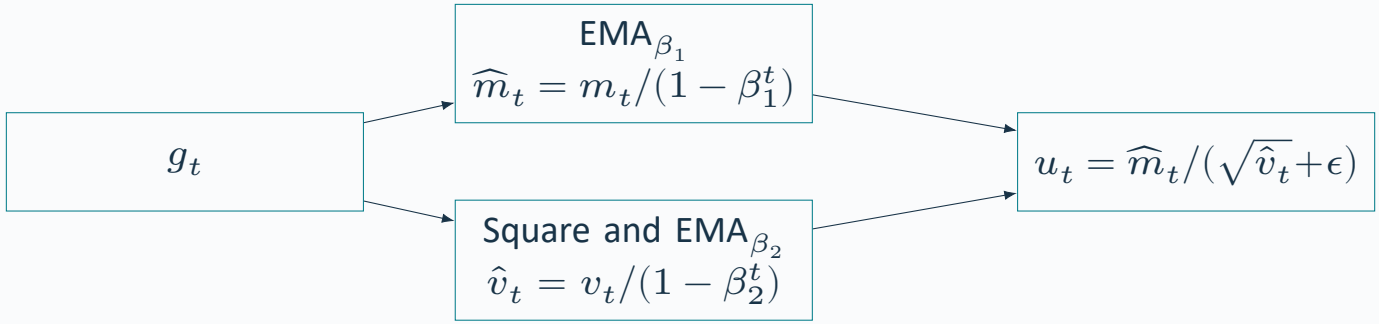
# AdamW: the second moment filters squared input

$$q_t = g_t^{\odot 2}, \qquad v_t = \beta_2 v_{t-1} + (1-\beta_2) q_t$$
$$\frac{V(z)}{Q(z)} = \frac{1-\beta_2}{1-\beta_2 z^{-1}}$$

This is a transfer from $q_t$ to $v_t$. The preceding map $g_t \mapsto q_t$ remains nonlinear.

**Derived in this presentation**. Kingma and Ba (2014 public; ICLR 2015); exact block decomposition

# AdamW: all mechanisms stay in the model




$$W_{t+1} = (1-\eta_t\lambda)W_t - \eta_t u_t$$

The schedule $\eta_t$, epsilon, bias corrections, and decay are all explicit.

Kingma and Ba (2014 public; ICLR 2015); Loshchilov and Hutter (2017 public; ICLR 2019)

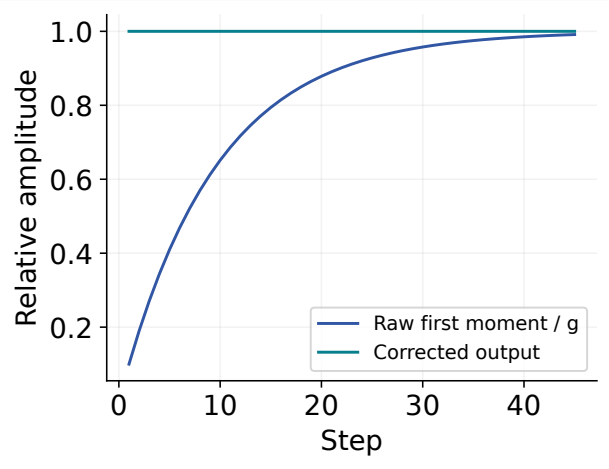
# AdamW has an exact constant-input response

$$g_t = g \quad \Longrightarrow \quad \widehat{m}_t = g, \quad \widehat{v}_t = g^{\odot 2}$$

$$u_t = \frac{g}{|g| + \epsilon}$$

For constant $d, \eta$ and $d \neq 1$:

$$W_{t+1} = d^t W_1 - \eta \frac{1 - d^t}{1 - d} u.$$




**Derived in this presentation**. Exact algebra with $m_0 = v_0 = 0$

# Muon: the exact Nesterov-style filter

$$B_t = \mu B_{t-1} + (1-\mu)G_t, \qquad Z_t = (1-\mu)G_t + \mu B_t$$

$$\boxed{\frac{Z(z)}{G(z)} = (1-\mu)\frac{1+\mu-\mu z^{-1}}{1-\mu z^{-1}}}$$

The unnormalized convention $\overline{B}_t = \mu\overline{B}_{t-1} + G_t$, $\overline{Z}_t = G_t + \mu\overline{B}_t$ removes the leading factor $1-\mu$.

**Derived in this presentation**. Jordan and contributors (Muon code, f90a42b); implementation-pinned derivation

# Muon: the finite nonlinear operator

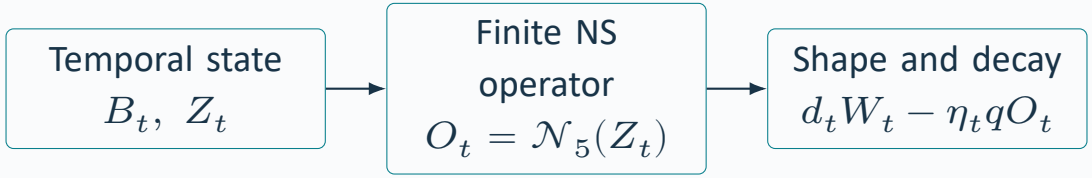
$$\mathcal{N}_5(Z) = \mathsf{untranspose}(X_5), \qquad X_0 = \frac{\mathsf{transposeIfTall}(Z)}{\|Z\|_F + \epsilon_{\mathsf{NS}}}$$

$$X_{k+1} = aX_k + bX_kX_k^\top X_k + c(X_kX_k^\top)^2X_k$$

- $k = 0, \dots, 4$ and $(a, b, c) = (3.4445, -4.775, 2.0315)$.
- The published code casts to bfloat16. Our analytical simulations use real-arithmetic formulas in float64.
- Finite precision adds rounding behavior beyond these exact algebraic recurrences.

Jordan and contributors (Muon code, f90a42b); PyTorch contributors (PyTorch Muon implementation)

# MuonW: the complete parameter loop




$$q = \sqrt{\max(1, m/n)}, \quad d_t = 1 - \eta_t \lambda$$

$$G_t = \nabla_W f(W_t), \qquad W \in \mathbb{R}^{m \times n}$$

The pinned shape scaling is one convention. Other implementations use RMS matching.

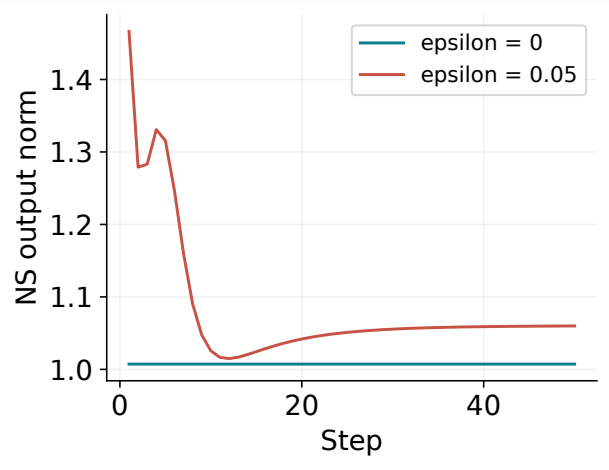
Jordan and contributors (Muon code, f90a42b); PyTorch contributors (PyTorch Muon implementation)

# Muon: constant-input response needs epsilon care

$$G_t = G,$$
$$B_t = (1-\mu^t)G,$$
$$Z_t = (1-\mu^{t+1})G.$$
$$X_{0,t} = \frac{G}{\|G\|_F + \epsilon_{\text{NS}}/(1-\mu^{t+1})}$$

$\epsilon_{\text{NS}} = 0$, $G \neq 0$: exactly fixed NS output.

$\epsilon_{\text{NS}} > 0$: generally time-dependent output.




**Derived in this presentation**. Exact recurrence, $B_0 = 0$; synthetic epsilon stress test

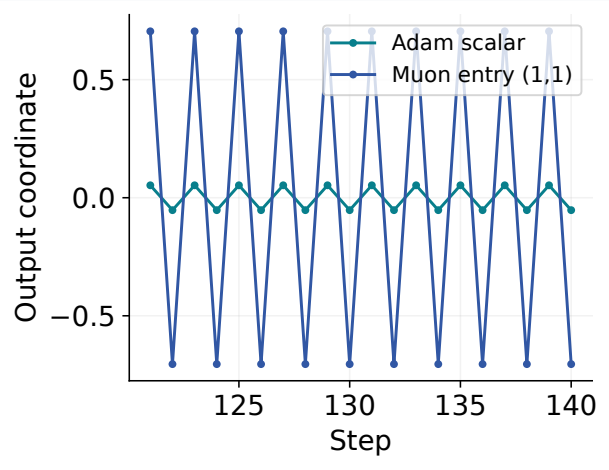
# AdamW and Muon under alternating gradients

$$g_t = (-1)^{t-1} g$$

$$u^*_{t,\text{Adam}} = (-1)^{t-1} \frac{1-\beta_1}{1+\beta_1} \frac{g}{|g|+\epsilon}$$

$$Z^*_{t,\text{Muon}} = (-1)^{t-1} \frac{(1-\mu)(1+2\mu)}{1+\mu} G$$

Muon output is $\mathcal{N}_5(Z^*_t)$, including epsilon.




**Derived in this presentation**. Exact asymptotic open-loop responses, fixed hyperparameters

# EMA has an exact periodic steady state

$$s_t = \beta s_{t-1} + (1-\beta) q_t, \quad q_{t+p} = q_t, \quad 0 \le \beta < 1$$

$$\boxed{s_t^* = \frac{1-\beta}{1-\beta^p} \sum_{j=0}^{p-1} \beta^j q_{t-j}}$$

$$s_t - s_t^* = \beta^t (s_0 - s_0^*)$$

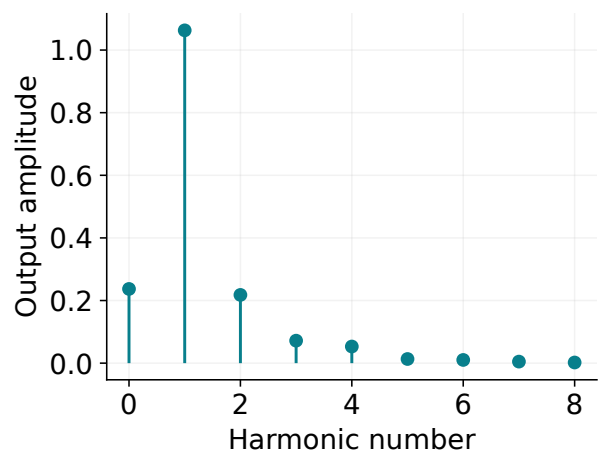
Apply this separately to $g_t$ and $g_t^{\odot 2}$. Adam bias corrections tend to one.

**Derived in this presentation**. Geometric-series proof in Appendix A3

# Nonlinear frequency analysis uses periodic outputs

$$\widehat{U}_k = \frac{1}{p}\sum_{t=0}^{p-1} U_t^* e^{-2\pi jkt/p}$$

- The output can contain new harmonics.
- Frequency content depends on amplitude and offset.
- Plot uses scalar Adam driven by $0.3 + \cos(2\pi t/64)$.




**Derived in this presentation**. Synthetic open-loop probe: $\beta_1 = .8$, $\beta_2 = .9$, $\epsilon = .03$

# Bounded inputs give qualified bounded outputs

$$\|g_t\|_\infty \le G, \quad m_0 = v_0 = 0, \quad \epsilon > 0 \implies \|u_t^{\mathsf{Adam}}\|_\infty \le G/\epsilon$$

- Muon: $\|X_0\|_F \le 1$ and five continuous polynomial steps give a finite dimension-dependent bound.
- Update bounds do not bound an unconstrained parameter integrator.
- Incremental bounds need a specified state domain and norm. They are not implied by BIBO.

**Derived in this presentation**. Bounds for the specified real-arithmetic operators

# A static sector constraint for Adam normalization

$$u = \frac{m}{\sqrt{v} + \epsilon} = a(v)m, \quad v \geq 0, \quad 0 < a(v) \leq 1/\epsilon$$

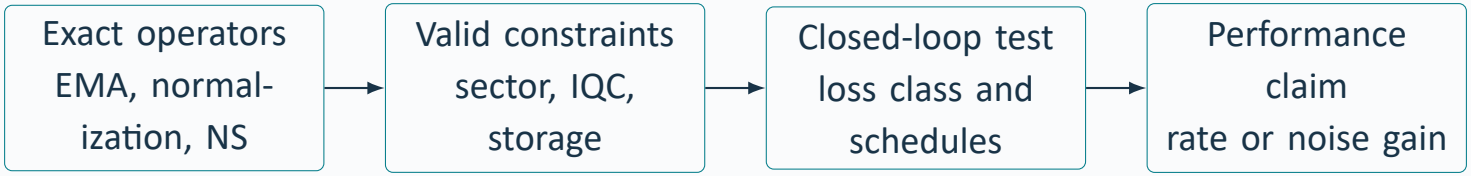
$$\boxed{u\left(\frac{m}{\epsilon} - u\right) = a(v)\left(\frac{1}{\epsilon} - a(v)\right)m^2 \geq 0}$$

$$\begin{bmatrix} m \\ u \end{bmatrix}^\top \begin{bmatrix} 0 & 1/(2\epsilon) \\ 1/(2\epsilon) & -1 \end{bmatrix} \begin{bmatrix} m \\ u \end{bmatrix} \geq 0$$

Use corrected moments as $m, v$ when applying this to Adam. This is an origin sector, not an incremental sector across different $v$.

**Derived in this presentation**. Pointwise proof shown in full

# A stability certificate needs the whole interconnection




- Published IQC methods supply a framework under explicit objective assumptions.
- The static sector above is one ingredient, not a completed AdamW or Muon theorem.
- Open problem: useful dynamic constraints for the complete modern optimizer loop.

**Research question**. Lessard, Recht, and Packard (2014 public; SIAM 2016); Hu and Lessard (ICML 2017)

# Case study: Muon to Hyperball

- Hyperball wraps a base optimizer with fixed matrix radius and normalized trial steps.
- Wen et al.: public June 2026, version 2 dated 25 October 2026.
- This chapter reconstructs its update from an output-regulation objective.
- The reconstruction is a design interpretation. Its radial guarantee alone does not establish training acceleration.

**Control-theoretic reinterpretation**. Wen, Dang, Lyu, Ma, and Liang (2026 preprint, v2); Xiao et al. (2026 preprint)

# Muon shapes the input to parameter dynamics

$$W_{t+1} = d_t W_t - \eta_t q O_t, \qquad O_t = \mathcal{N}_5(Z_t)$$

| Quantity | Role |
|---|---|
| $B_t$ | Temporal controller state |
| $O_t$ | Matrix update after finite spectral shaping |
| $W_t$ | Parameter state |
| $r_t = \|W_t\|_F^2$ | Candidate controlled output |

**Control-theoretic reinterpretation**. Pinned MuonW recurrence, interpreted as a control system

# The radial state has an exact recurrence

$$\boxed{\begin{aligned} r_{t+1} = d_t^2 r_t \\ - 2 d_t \eta_t q \langle W_t, O_t \rangle_F \\ + \eta_t^2 q^2 \|O_t\|_F^2. \end{aligned}}$$

This is the squared norm of an affine update, with no Taylor expansion.

**Derived in this presentation**. Exact identity for every real matrix of compatible shape

# Equal input norms can produce different radii

$$\|W_t\|_F = R, \quad \|O_t\|_F = 1, \quad d_t = q = 1$$

$$r_{t+1} = R^2 - 2\eta R \cos\theta_t + \eta^2$$

| Alignment | Next squared radius |
|---|---|
| $O_t = W_t/R$ | $(R-\eta)^2$ |
| $\langle W_t, O_t \rangle_F = 0$ | $R^2 + \eta^2$ |
| $O_t = -W_t/R$ | $(R+\eta)^2$ |

**Derived in this presentation**. Explicit simplified examples of the exact norm identity

# The desired output response is a design choice

$$R = \|W_1\|_F > 0, \qquad e_t = \|W_t\|_F^2 - R^2$$

$$\boxed{e_{t+1} = 0}$$

- Deadbeat output regulation: remove the output error in one step.
- Choose the weight radius as a controlled quantity.
- This objective is an added specification. Muon does not uniquely imply it.

**Control-theoretic reinterpretation**. Output-regulation specification used in this presentation

# Input amplitude regulation

$$v_t = \frac{O_t}{\|O_t\|_F}, \qquad \widetilde{W}_{t+1} = W_t - \eta_t R v_t$$

$$\|\widetilde{W}_{t+1} - W_t\|_F = \eta_t R$$

- The normalized direction retains the finite NS output direction.
- Assume $O_t e 0$. A zero-update implementation needs an explicit policy.
- On the sphere, $0 \leq \eta_t < 1$ guarantees a nonzero trial point.

**Derived in this presentation**. Exact input normalization, with stated domain

# Scalar feedback gain from the target response

$$W_{t+1} = k_t \widetilde{W}_{t+1}, \qquad k_t > 0$$

$$\|W_{t+1}\|_F^2 = k_t^2 \|\widetilde{W}_{t+1}\|_F^2 = R^2$$

$$\boxed{k_t = \frac{R}{\|\widetilde{W}_{t+1}\|_F}}$$

Unique positive gain for a nonzero trial state and a positive reference radius.

**Derived in this presentation**. Exact feedback synthesis

# The feedback law recovers Hyperball

$$W_{t+1} = R\frac{W_t - \eta_t R\, O_t/\|O_t\|_F}{\|W_t - \eta_t R\, O_t/\|O_t\|_F\|_F}$$




Positive shape factors cancel in $O_t/\|O_t\|_F$. Base momentum and finite NS dynamics remain.

**Control-theoretic reinterpretation**. Wen, Dang, Lyu, Ma, and Liang (2026 preprint, v2); feedback derivation shown here

# One-step regulation has a precise guarantee

$$\|W_{t+1}\|_F^2 = \frac{R^2}{\|\widetilde{W}_{t+1}\|_F^2}\|\widetilde{W}_{t+1}\|_F^2 = R^2$$

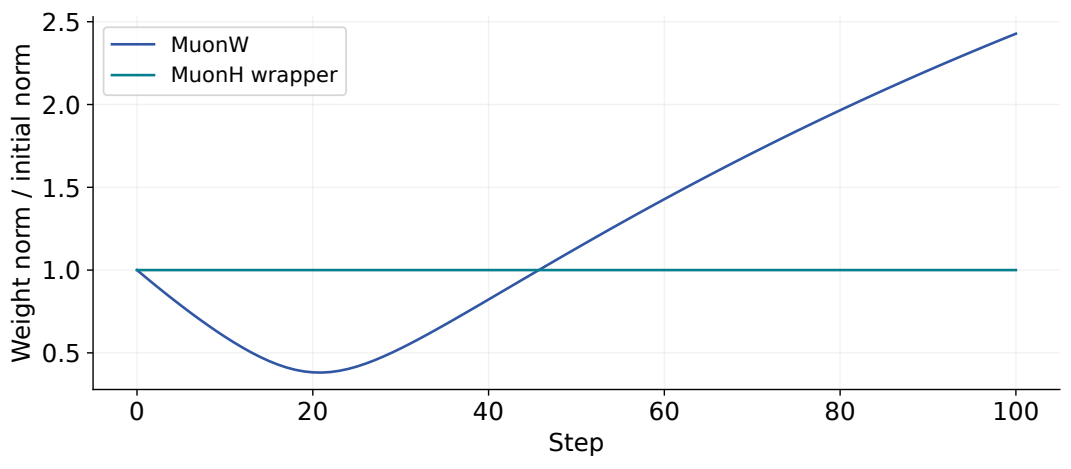
$$e_{t+1} = 0$$

$$\|W_t\|_F = R \quad \Longrightarrow \quad \|W_{t+1} - W_t\|_F \leq 2\eta_t R$$

The last bound follows from the triangle and reverse-triangle inequalities. Neither equation proves a decrease in $f$.

**Derived in this presentation**. Exact proof and displacement bound

# MuonW and MuonH have different radial responses




Synthetic open-loop probe: fixed $G = \mathrm{diag}(1, .12)$, $\mu = .95$, five NS steps, $\eta = .04$, $q = 1$, $\lambda = .2$ for MuonW. MuonH uses the same base output and no weight decay.

**Illustrative exact system**. Reproducible synthetic open-loop simulation

# Radial regulation can coexist with a two-cycle

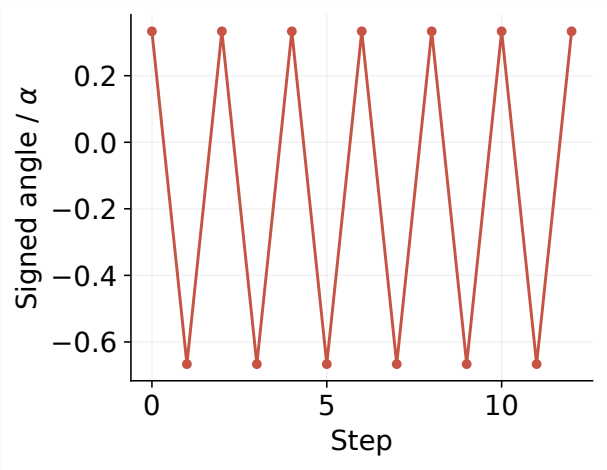
$$f(w) = -w_1/\|w\|, \quad \|w\| = 1$$

$$w = (\cos\phi, \sin\phi), \quad \alpha = \arctan\eta$$

$$\phi_{t+1} = \phi_t - \alpha\,\mathrm{sign}(\phi_t)$$

Normalized gradient, no momentum or NS.

$\phi_0 = \alpha/3$ gives $\alpha/3 \leftrightarrow -2\alpha/3$.




**Derived in this presentation**. Exact illustrative closed loop; $\eta = .35$; proof in Appendix A5

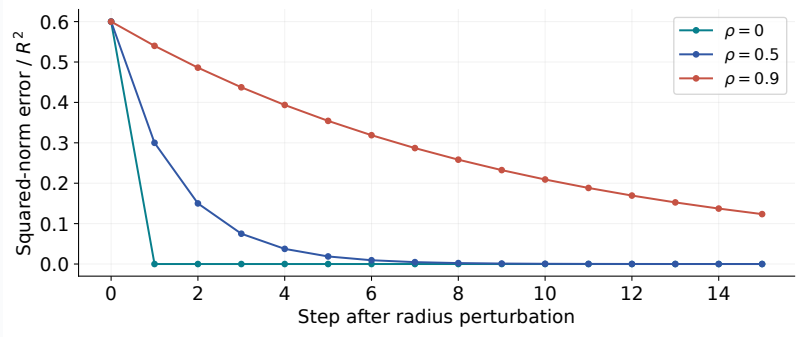
# A designed radial pole generalizes the wrapper

$$e_{t+1} = \rho e_t, \qquad 0 \le \rho < 1$$

$$k_t = \frac{\sqrt{R^2 + \rho(r_t - R^2)}}{\|\widetilde{W}_{t+1}\|_F}$$

$\rho = 0$: Hyperball.

If $r_1 = R^2$, all choices preserve the same radius.




**Derived in this presentation**. Exact feedback construction; nonzero candidate required

# From optimizer analysis to controller synthesis

1. **Optimizers are dynamical systems.**
   Internal memory and nonlinear maps determine their responses.
2. **Exact analysis gives qualified statements.**
   Open-loop responses, bounded updates, and radial regulation have different scopes.
3. **Controller synthesis starts with a specification.**
   Hyperball follows from a chosen norm target and scalar feedback structure.

Can modern optimizers be synthesized from desired nonlinear closed-loop behavior?

**Research question**. Lessard, Recht, and Packard (2014 public; SIAM 2016); Hu and Lessard (ACC 2017); Wen, Dang, Lyu, Ma, and Liang (2026 preprint, v2)

# A1. Timing, conventions, and domain assumptions

| Item | Convention |
|---|---|
| Time | $t \geq 1$: read $W_t$, form $g_t$, update $s_t$, write $W_{t+1}$ |
| Initial buffers | Adam $m_0 = v_0 = 0$. Muon $B_0 = 0$ |
| Fixed-filter responses | $0 \leq \beta_1, \beta_2, \mu < 1$, fixed hyperparameters |
| Numerical model | Real-arithmetic recurrences, simulated in float64 |
| Normalization | Adam $\epsilon > 0$. Muon additive $\epsilon_{\text{NS}} = 10^{-7}$ |
| Hyperball | $R > 0$, nonzero update and candidate. Zero update may be skipped |

Jordan and contributors (Muon code, f90a42b); Wen, Dang, Lyu, Ma, and Liang (2026 preprint, v2)

# A2. Heavy-ball stability and Muon conventions

$$p(z) = z^2 - az + \beta, \qquad a = 1 + \beta - \eta h$$

$$1 - a + \beta > 0, \quad 1 + a + \beta > 0, \quad 1 - \beta > 0$$

$$0 \le \beta < 1 : \quad 0 < \eta h < 2(1 + \beta)$$

$$B_t = (1 - \mu)\overline{B}_t, \qquad Z_t = (1 - \mu)\overline{Z}_t$$

The last identities hold for zero buffers. Under $Z/(\|Z\|_F + \epsilon_{\mathsf{NS}})$, a scalar rescaling changes the effective epsilon.

**Derived in this presentation**. Algebraic derivations for the stated scalar and matrix systems

# A3. Periodic EMA and alternating-input proofs

$$s_t = \beta^p s_{t-p} + (1-\beta)\sum_{j=0}^{p-1} \beta^j q_{t-j}$$

$$s_t^* = s_{t-p}^* \implies (1-\beta^p)s_t^* = (1-\beta)\sum_{j=0}^{p-1} \beta^j q_{t-j}$$

$$q_t = (-1)^{t-1} q: \quad s_t^* = (-1)^{t-1}\frac{1-\beta}{1+\beta} q$$

$$Z_t^* = (1-\mu)G_t + \mu B_t^* = (-1)^{t-1}\frac{(1-\mu)(1+2\mu)}{1+\mu} G$$

**Derived in this presentation**. Exact geometric-series derivations

# A4. Bounded outputs and a displacement bound

$$K_0 = 1, \qquad K_{k+1} = |a|K_k + |b|K_k^3 + |c|K_k^5$$

$$\|X_k\|_F \le K_k \implies \|\mathcal{N}_5(Z)\|_F \le K_5 < \infty$$

$$\begin{aligned}\|W_{t+1} - W_t\|_F &\le \|W_{t+1} - \widetilde{W}_{t+1}\|_F + \eta_t R \\ &= \left|R - \|\widetilde{W}_{t+1}\|_F\right| + \eta_t R \le 2\eta_t R.\end{aligned}$$

The NS bound is deliberately loose. The displacement proof assumes $\|W_t\|_F = R$ and a nonzero candidate.

**Derived in this presentation**. Direct norm inequalities

# A5. The two-cycle is an exact closed-loop example

$$f(w) = -\frac{w_1}{\|w\|}, \quad \nabla f(w) = \sin\phi\, e_\phi, \quad e_\phi = (-\sin\phi, \cos\phi) \quad (\|w\| = 1)$$

$$v = \text{sign}(\phi) e_\phi \quad \text{for } 0 < |\phi| < \pi/2$$

$$\frac{w - \eta v}{\|w - \eta v\|} = \cos\alpha\, w - \text{sign}(\phi) \sin\alpha\, e_\phi, \quad \alpha = \arctan\eta$$

$$\phi_0 = \alpha/3 \implies \phi_1 = -2\alpha/3, \quad \phi_2 = \alpha/3$$

The base direction is the normalized gradient. This is not the full momentum-plus-NS MuonH controller.

**Derived in this presentation**. Exact counterexample, smooth objective on $\mathbb{R}^2 \setminus \{0\}$

# A6. A sufficient static-IQC certificate

$$\xi_{t+1} = A\xi_t + Bu_t, \quad y_t = C\xi_t + Du_t, \quad u_t = \Delta(y_t)$$

$$\begin{bmatrix} y_t \\ u_t \end{bmatrix}^\top M \begin{bmatrix} y_t \\ u_t \end{bmatrix} \geq 0$$

$$\begin{bmatrix} A & B \end{bmatrix}^\top P \begin{bmatrix} A & B \end{bmatrix} - \begin{bmatrix} \rho^2 P & 0 \\ 0 & 0 \end{bmatrix} + \lambda \begin{bmatrix} C & D \\ 0 & I \end{bmatrix}^\top M \begin{bmatrix} C & D \\ 0 & I \end{bmatrix} \preceq 0$$

$P \succ 0, \lambda \geq 0, 0 < \rho < 1$ imply $V_{t+1} \leq \rho^2 V_t$, assuming a well-posed, equilibrium-shifted interconnection and a valid pointwise constraint. Dynamic IQCs need augmented states and appropriate hard or weighted conditions.

**Derived in this presentation**. Lessard, Recht, and Packard (2014 public; SIAM 2016); static certificate specialized here

# A7. Selected references: optimization and control


1. Kingma and Ba. *Adam: A Method for Stochastic Optimization*. 2014 preprint, ICLR 2015. arXiv:1412.6980.
2. Loshchilov and Hutter. *Decoupled Weight Decay Regularization*. 2017 preprint, ICLR 2019. arXiv:1711.05101.
3. Lessard, Recht, and Packard. *Analysis and Design of Optimization Algorithms via IQCs*. 2014 preprint, SIAM J. Optim. 2016. arXiv:1408.3595.
4. Hu and Lessard. *Control Interpretations for First-Order Optimization Methods*. ACC 2017. arXiv:1703.01670.
5. Hu and Lessard. *Dissipativity Theory for Nesterov's Accelerated Method*. ICML 2017. PMLR 70.
6. Chakrabarti and Chopra. *A Control Theoretic Framework for Adaptive Gradient Optimizers in Machine Learning*. 2022 preprint, v2 2023. arXiv:2206.02034.
7. Chen et al. *Accelerated optimization in deep learning with a proportional-integral-derivative controller*. Nature Communications 15, 10263, 2024. DOI.


Full bibliographic metadata: references.bib and source_ledger.csv

# A8. Selected references: matrix methods and Hyperball


1. Gupta, Koren, and Singer. *Shampoo: Preconditioned Stochastic Tensor Optimization*. ICML 2018. PMLR 80.
2. Vyas et al. *SOAP: Improving and Stabilizing Shampoo using Adam for Language Modeling*. 2024 preprint, ICLR 2025. arXiv:2409.11321.
3. Jordan. *Muon: An optimizer for hidden layers in neural networks*. 8 December 2024. Author write-up. Implementation pinned to `f90a42b`.
4. Wen et al. *Fantastic Pretraining Optimizers and Where to Find Them II: Hyperball Optimization*. 15 June 2026, v2 25 October 2026. arXiv:2606.16899v2.
5. Xiao et al. *Hyperball May Not Be a Free Lunch*. 24 July 2026. arXiv:2607.22444.
6. Khona et al. *SOAP, Muon, and Beyond: Pushing LLM Pretraining Scales*. 13 July 2026. arXiv:2607.20548.


Version and venue status checked on 9 November 2026